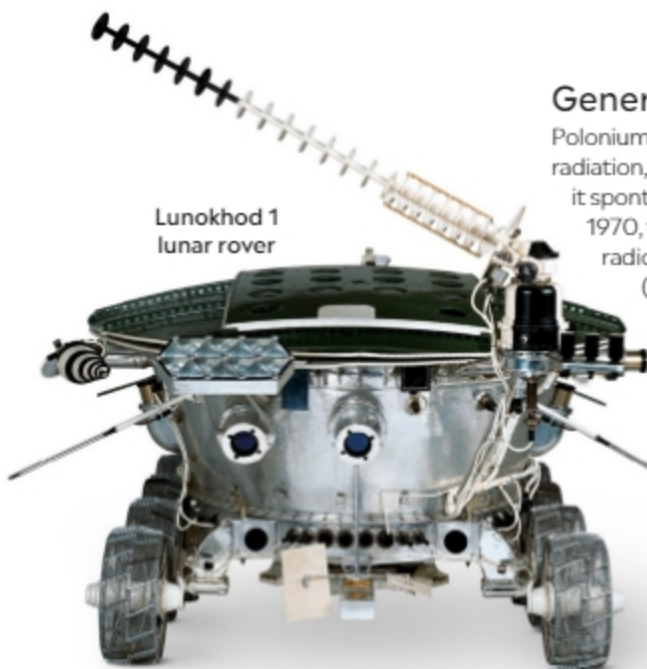




Lunokhod 1
lunar rover

Generating heat

Polonium emits a lot of radiation, and because of this it spontaneously heats up. In 1970, this heat powered a radioisotopic heater unit (RHU) on Lunokhod 1, a remotely operated rover placed on the moon by the Russians. The RHU kept the rover's instruments at the correct operating temperature at night, when the temperature plummeted to -279.4°F (-173°C).

Polonium

84
Po

This element was named after Poland, the birth country of its discoverer—Marie Curie. This silvery-gray metal is rare and highly radioactive, making it very dangerous to humans.

Livermorium

116

Lv

Named in 2012, livermorium is an artificial, super-heavy element that was made by colliding curium and calcium atoms in a machine called a particle accelerator. It was discovered in partnership by Russian scientists at the Joint Institute for Nuclear Research (JINR) in Dubna, Russia, and American scientists at the Lawrence Livermore National Laboratory in California. Livermorium is named after this American laboratory. Little is known about this element, but scientists think it's likely to have similar properties to polonium.



Vulcanization

Adding sulfur to rubber hardens the rubber and makes it flexible and more durable. This process is called vulcanization. Vulcanized rubber is up to 10 times stronger than natural rubber. Vehicle tires and rubber hoses are made of vulcanized rubber.



Acid rain

Burning fossil fuels releases sulfur and nitrogen compounds into the air. These react with oxygen and water to form acids, which then fall as rain. This acid rain can corrode stone structures, especially statues and buildings made from limestone and marble.

Hot springs in the Danakil
Depression, Ethiopia

